

Thermodynamics

Lecture 2B:

- Energy Transfer, General Energy Analysis

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After this lecture

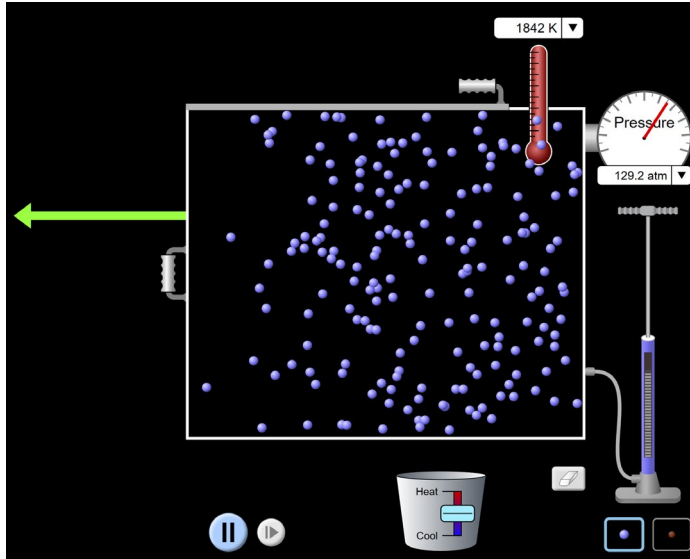
- Define the concept of work, including electrical and several forms of mechanical work.
- Understand the first law of thermodynamics, energy balances and mechanisms of energy transfer.
- Solving simple single-answer problems using the first law.

Learning objectives

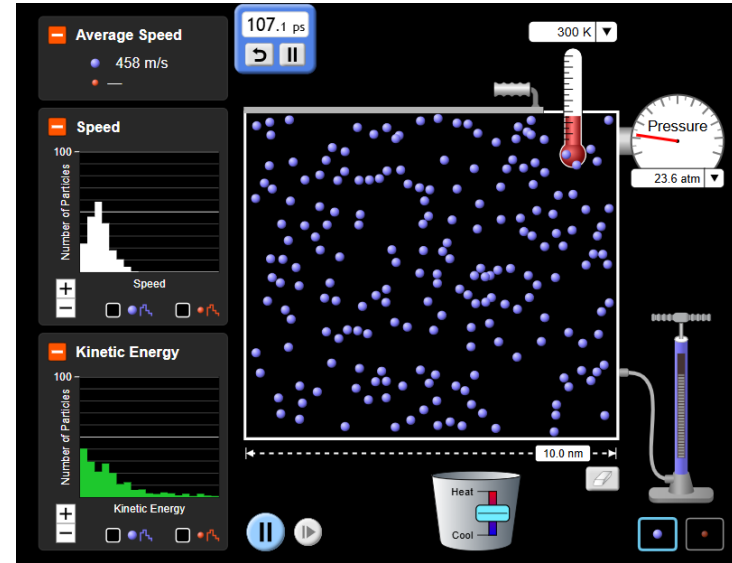
What you will be able to do after this course:

- I. *Knowledge:* Write the first and second laws of thermodynamics.**
- II. *Comprehension:* Describe the first and second laws in the student's own language.**
- III. *Application:* Solve simple single-answer problems using the first law.**
- IV. *Analysis:* Solve problems requiring both the first and second laws.
- V. *Synthesis:* Develop schemes to improve the efficiency or performance of thermodynamic systems, e.g., compressors, turbines, heat pumps, etc.

Recap demonstrations:



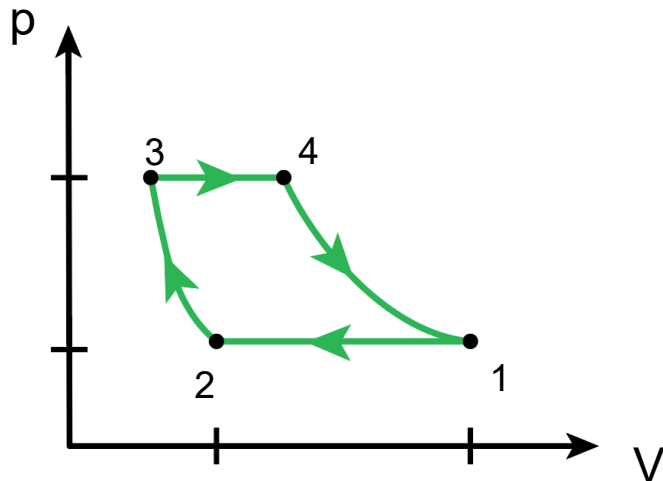
Non-quasistatic process



Measuring state variables

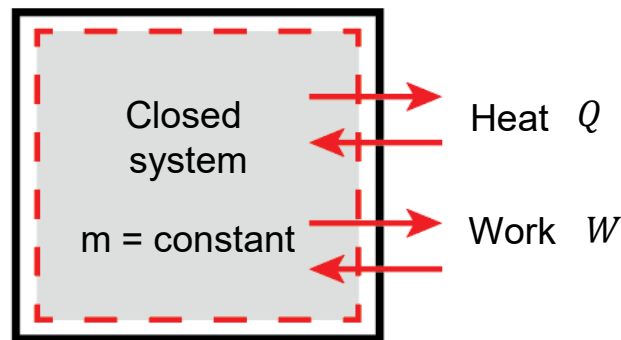
Recap Lecture 2A

- State properties ρ, T, p
- Internal energy U



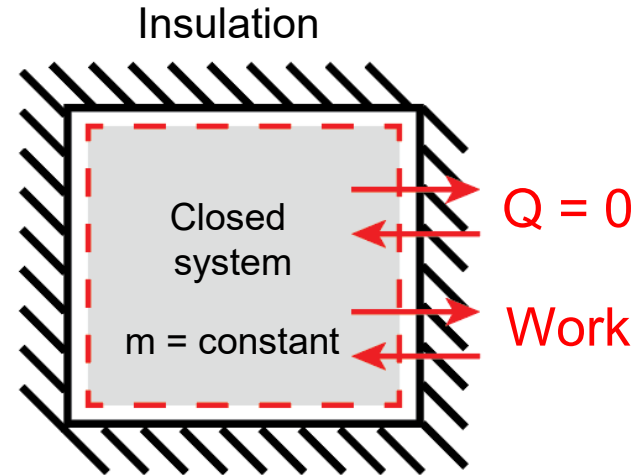
p-V diagram of a thermodynamic cycle

$$\Delta E = \Delta U + \Delta KE + \Delta PE + \dots$$



Adiabatic process

- A process where a system does not exchange heat with the surroundings.
- *Adiabatos*: not to be passed.



Energy transfer by work

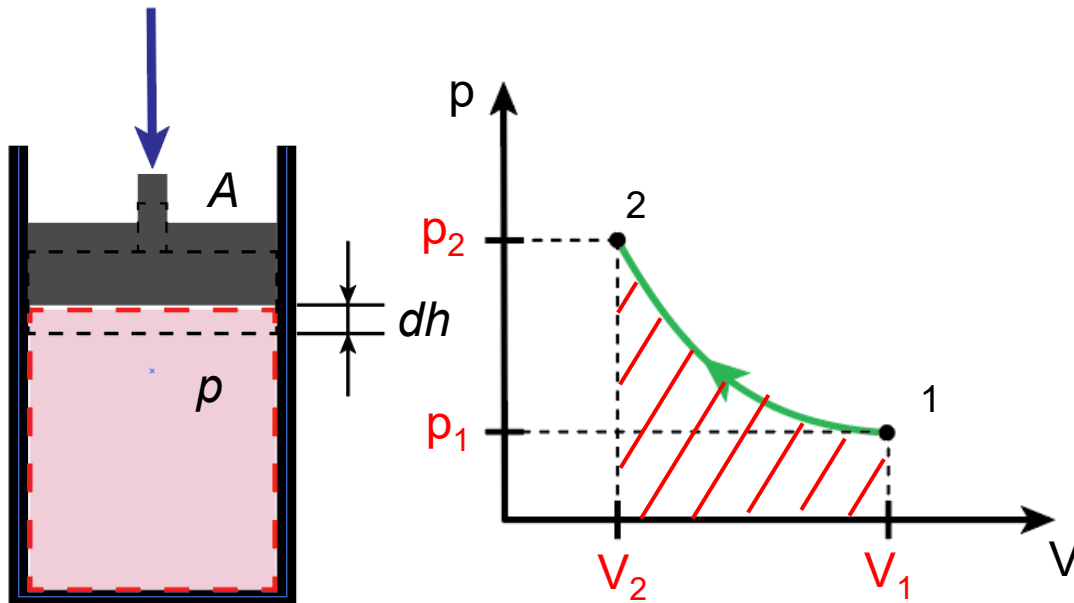
- Work transfer on a closed system

$$dW = Fdh$$

$$dW = pAdh$$

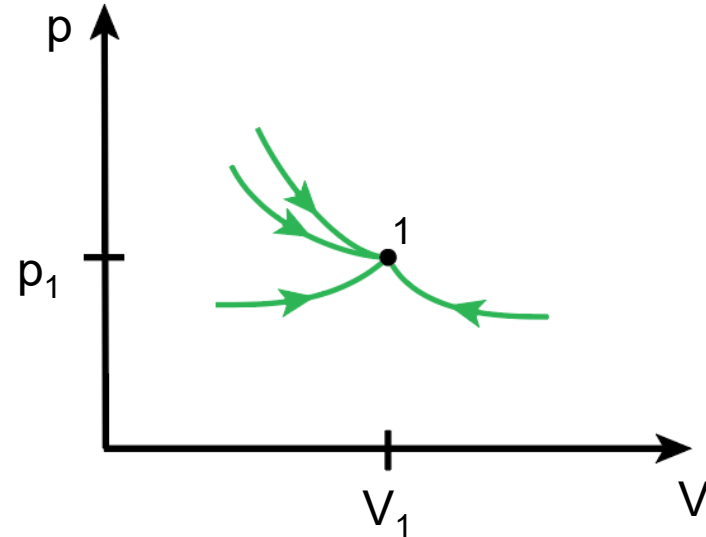
$$dW = pdV$$

$$W = \int_1^2 pdV$$



Thermodynamic state properties

- The properties of a state do not depend on the process that it was reached.
- State properties are point functions.

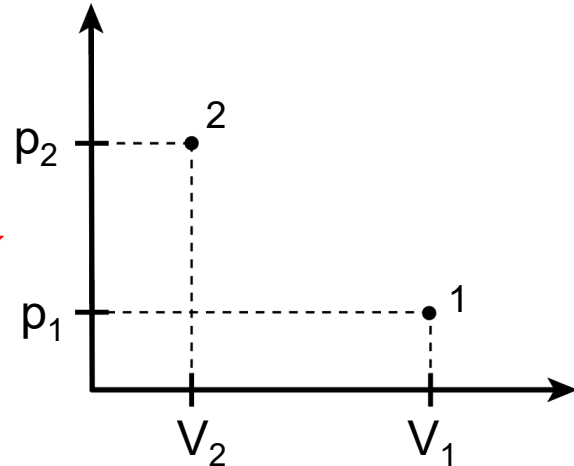


Energy transfer: Heat and work transfer

- Boundary phenomena: heat and work transfer occur at boundary of system.
- Are associated with a *process* not a state.
- Q and W depend on the *path* followed during the process, they are path functions

Path function: Inexact differentials $\delta Q; \delta W$

Point functions: Exact differentials dV



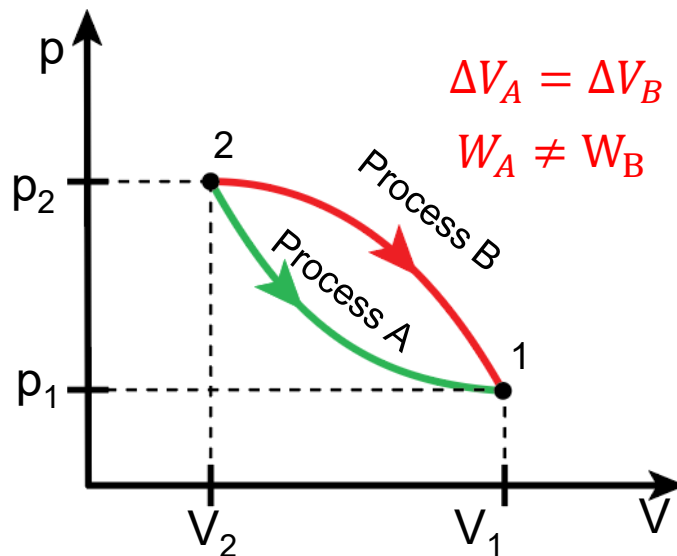
Energy Transfer: Heat and work transfer

- Q and W are path dependent.

$$\int_1^2 dV = V_2 - V_1$$

$$\int_1^2 \delta W = W_{12} \neq W_2 - W_1$$

$$W = \int_1^2 p dV$$

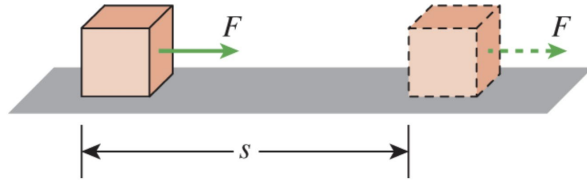


Types of mechanical work

- Linear mechanical work

Force F

Distance s



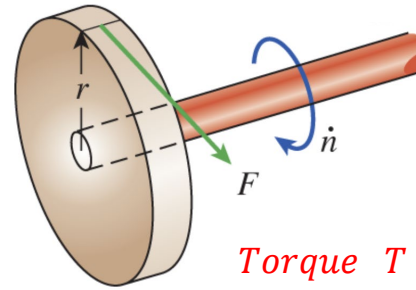
$$W = Fs$$

$$\dot{W} = FV$$

$$W = \int_1^2 F ds$$

- Shaft work

Rotational speed $\dot{n}$



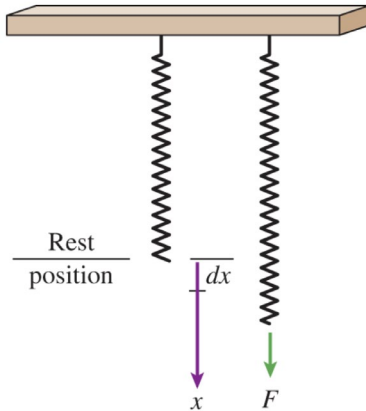
Torque $T = Fr$

$$W_{sh} = Fs = \left(\frac{T}{r}\right)(2\pi r n) = 2\pi n T$$

$$\dot{W}_{sh} = 2\pi \dot{n} T$$

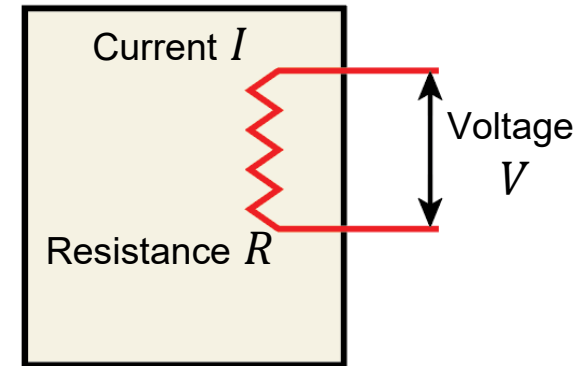
Types of work

- Spring work (mechanical)



$$F = kx$$
$$W = \int_1^2 F dx = \int_1^2 kx dx$$
$$W = \frac{1}{2}k (x_2^2 - x_1^2)$$

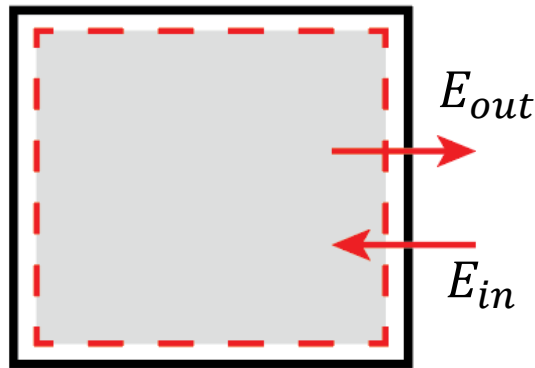
- Electrical work



$$\dot{W}_e = VI = I^2 R = \frac{V^2}{R}$$

Example: Electric Water Kettle

Conservation of energy principle

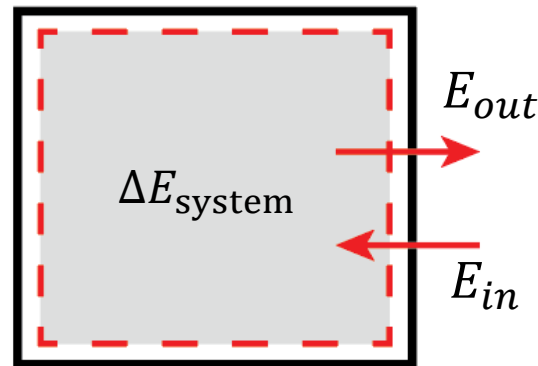


Energy balance

Conservation of energy principle

$$\left(\text{Total energy} \right)_{\text{into system}} - \left(\text{Total energy} \right)_{\text{out of system}} = \left(\text{Change in total energy of the system} \right)$$

$$E_{in} - E_{out} = \Delta E_{\text{system}}$$



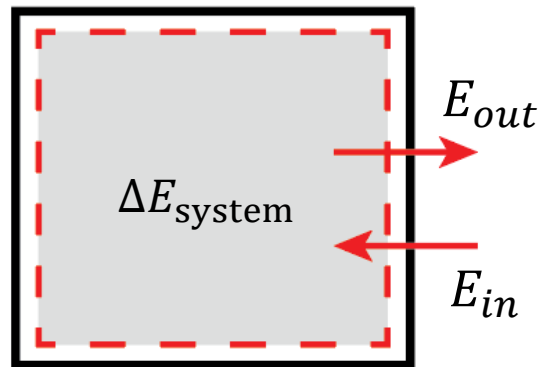
Energy change of a system

$$\Delta E_{\text{system}} = \Delta U + \Delta KE + \Delta PE + \dots$$

$$\Delta E_{\text{system}} = \Delta U$$

Stationary systems

$$\Delta KE = 0 \quad \Delta PE = 0$$



Energy transfer in/out of a closed system

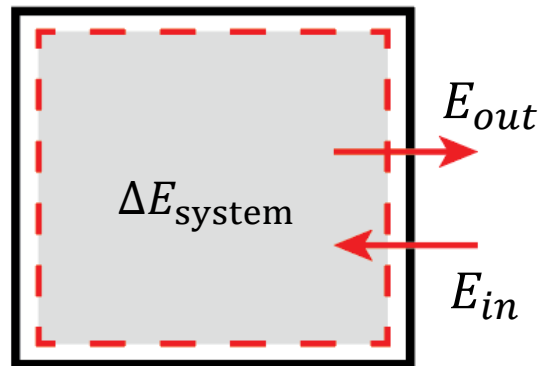
- Account for energy transfer with surroundings.

$$E_{in} - E_{out} = (Q_{in} - Q_{out}) + (W_{in} - W_{out})$$

Q $-W$

Heat in: $Q > 0$

Work out: $W > 0$

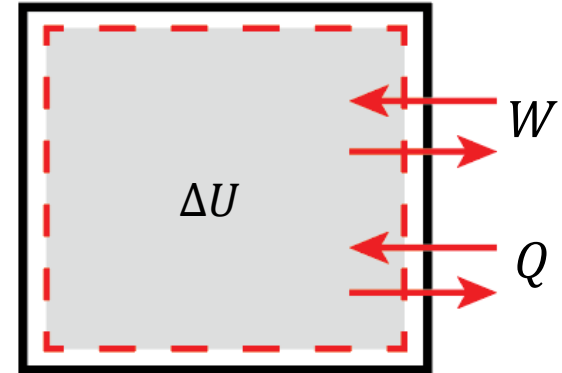


1st Law of thermodynamics: Closed system

$$E_{in} - E_{out} = \Delta E_{\text{system}}$$

$$\Delta U = Q - W$$

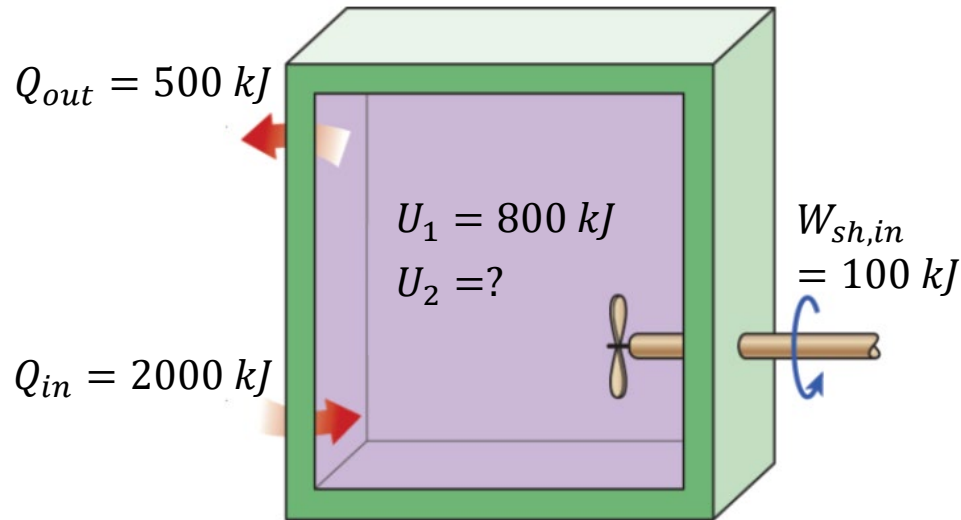
$$\sum \Delta U_i = \sum Q_i - \sum W_i$$



Group Work: in groups of 2; solve the following problem.



Given:



Question:

1. Determine U_2 of the air inside the box?

To Do:

- Define the thermodynamic system of interest.
- Write down an energy balance of the system.
- Go to pollev.com/claudiohail
- ... and answer question 1.

1st Law of thermodynamics: Example I

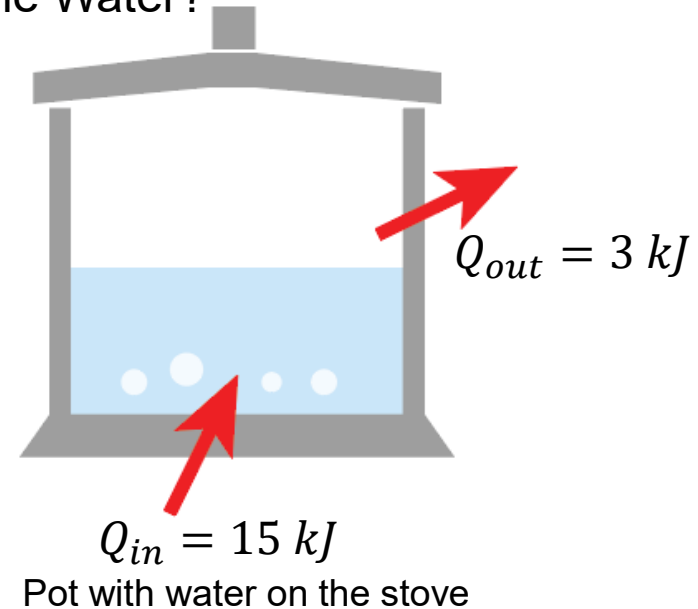
- Water is heated in a pot on a stove. Water experiences a heat transfer of $Q_{in} = 15$ kJ from the stove and loses $Q_{out} = 3$ kJ to the atmosphere via the pot. What is the change in internal energy of the Water?
 - Define thermodynamic system

$$\Delta U = Q - W$$

$$Q = Q_{in} - Q_{out}; W = 0;$$

$$\Delta U = \Delta U_{water} + \Delta U_{pot}$$

$$\Delta U = Q_{in} - Q_{out} = 12 \text{ kJ}$$



1st Law of thermodynamics: Example II

- Determine U_2 of the air inside the box.

- Define the system

$$\Delta U = Q - W$$

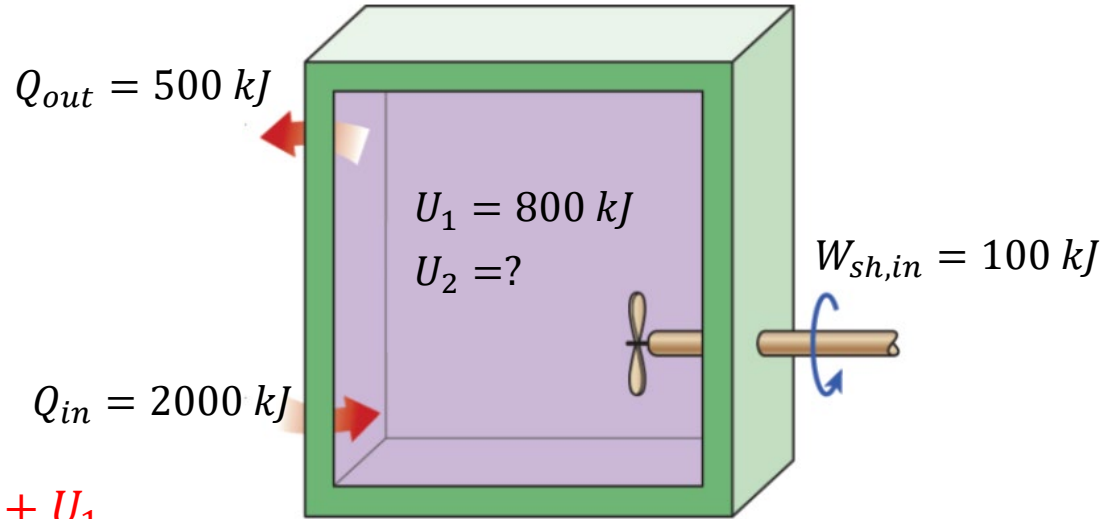
$$W = -100 \text{ kJ} ;$$

$$\Delta U = U_2 - U_1$$

$$Q = Q_{in} - Q_{out} = 1500 \text{ kJ} \quad Q_{in} = 2000 \text{ kJ}$$

$$U_2 = Q_{in} - Q_{out} + W_{shaft,in} + U_1$$

$$U_2 = Q_{in} - Q_{out} + W_{shaft,in} + U_1 = 2400 \text{ kJ}$$



Summary

- Discussed definitions of density, temperature and pressure.
- Understand the concept of energy and its different forms.
- Learn about the nature of internal energy.
- Concept of energy transfer by heat and work.

Next up

- Energy analysis of cycles
- Energy conversion efficiency